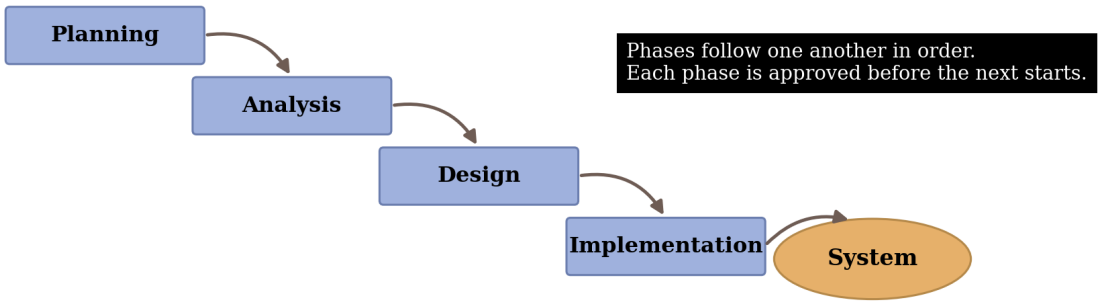
Disaster Alert & Resource Allocation System

1. Methodology: Waterfall



- Strict deadline: the system must work before the next disaster season.
- Clear requirements: alerts by area and severity, live resource inventory, dashboard.
- The 90-day plan is already split into four sequential phases.

2. Time Estimation

Phase	%	Days
Planning	15%	14
Analysis	20%	18
Design	35%	31
Implementation	30%	27
Total	**100%**	**90**

3. Case Study

Background: the Civil Defense and Emergency Management Authority needs to warn people quickly during disasters and send limited rescue and relief resources where they are needed most.

Problems identified

- Warnings reach people late or through few channels.
- Teams, shelters and supplies are assigned by phone and paper lists, causing wrong or duplicated deliveries.
- No live picture of affected areas, stock and unmet needs.
- Records are lost and manual counting causes errors.

Project objectives

- Send alerts by SMS, email and push within minutes.
- Keep a live inventory of ambulances, rescue teams, water, food, medicine and shelter beds.
- Suggest resource allocation by severity and population.
- Give decision makers a dashboard with maps and reports.

Feasibility

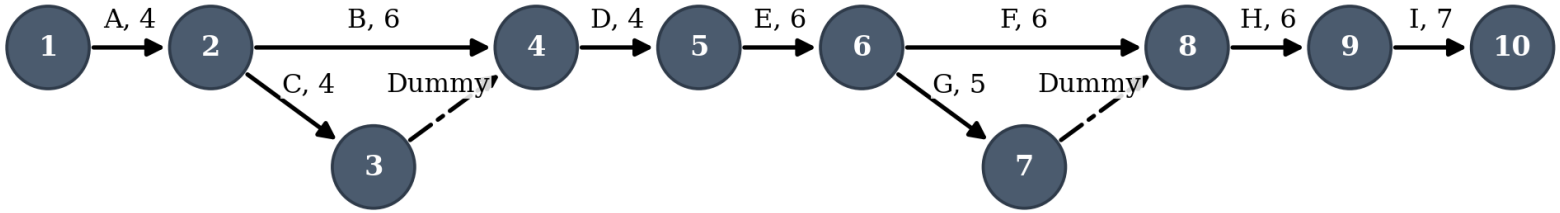
Type	Result
Technical	Feasible (web application, Leaflet maps, Node.js + PostgreSQL, alert gateway)
Operational	Feasible (live inventory, one dashboard, trained operators)
Economic	Feasible (cost $3,000 – $6,200, savings $8,000 – $15,000+ per year, payback within 1 year)
Legal	Feasible (privacy policy, role-based access, data-sharing agreements)
Schedule	Feasible (90 days)

Activity List

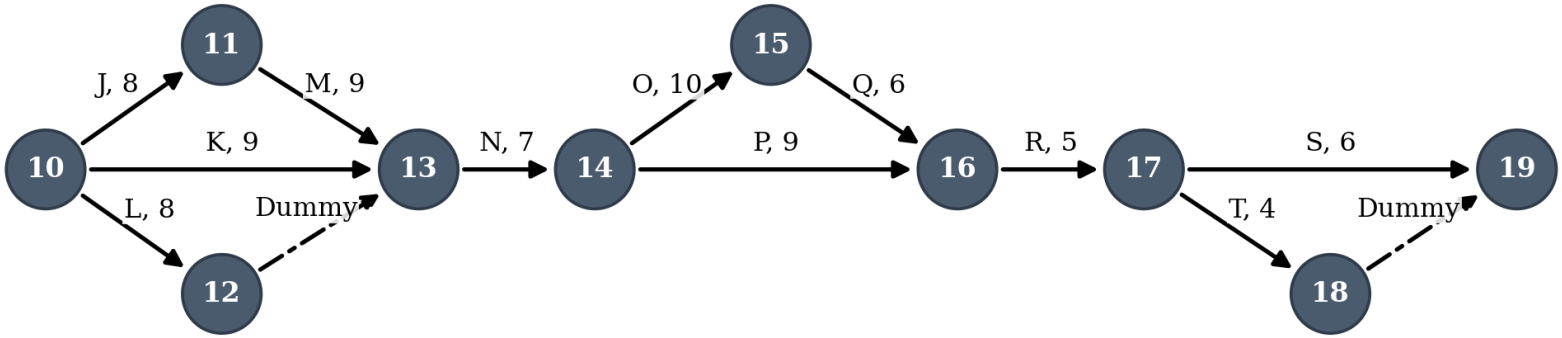
Activity	Description	Duration (Days)	Predecessor
A	Project initiation and scope	4	-
B	Feasibility study	6	A
C	Resource and risk plan	4	A
D	Plan approval	4	B, C
E	Requirements gathering	6	D
F	Process and data modelling	6	E
G	Security and legal requirements	5	E
H	Requirements sign-off	6	F, G
I	Architecture design	7	H
J	Database design	8	I
K	UI and map dashboard design	9	I
L	Alert and gateway design	8	I
M	Allocation logic design	9	J
N	Design review and approval	7	M, K, L
O	Backend development	10	N
P	Frontend development	9	N
Q	SMS, email, push and data-feed integration	6	O
R	Testing	5	P, Q
S	UAT and go-live	6	R
T	Training	4	R

4. PERT Chart

Part 1: Planning and Analysis (events 1–10)



Part 2: Design and Implementation (events 10–19, event 10 continues from Part 1)



Critical path: A → B → D → E → F → H → I → J → M → N → O → Q → R → S = **90 days**

5. Gantt Chart

